VEduTutorBench: a High-Quality Vietnamese Secondary-school Informatics AI Tutor Benchmark from Teacher-Authored Dialogues

Anh-Quan Nguyen-Duc[1,*] 26025008@vnu.edu.vn
Huu-Quynh Nguyen[2] nhquynh@cmcu.edu.vn
Thi-Mai Hoang[3] htmai@hnmu.edu.vn
Anh-Hoang Truong[2] tahoang@cmcu.edu.vn
Van-Vinh Nguyen[1] vinhnv@vnu.edu.vn
Chi-Chung Le[3] lcchung@hnmu.edu.cn
Hong-Cuong Do[3] dhcuong@hnmu.edu.vn
Khoi-Nguyen Duong[1] 23020127@vnu.edu.vn

[1]*VNU University of Engineering and Technology (UET)*, Hanoi, Vietnam; [2]*CMC University*, Hanoi, Vietnam
[3]*Hanoi Metropolitan University (HNMU)*, Hanoi, Vietnam; [*]Corresponding author: Anh-Quan Nguyen-Duc

***Abstract*---Evaluating a large language model as a tutor requires interpreting the learner's state, selecting an appropriate pedagogical move, and providing calibrated support, not merely checking answer correctness. We present VEduTutorBench, a Vietnamese Grades 6-9 Informatics AI-tutor benchmark built through a traceable pipeline from 1,050 teacher-authored dialogues. The pipeline audits the dialogues, establishes six pedagogical principles and six tutor capabilities, and converts eligible tutor turns into 2,028 context-conditioned candidates from 665 retained dialogue families. A grounded LLM scores the necessity of all six principles for each candidate; deterministic checks retain 1,400 candidates across 665 dialogue families. Our evaluation framework preserves native multi-turn roles and compares tutor generated and teacher-authored responses using four common criteria, three criteria per required principle, and criterion-level Win/Tie/Lose judgments. We evaluate three tutor configurations-two Gemini-based tutors and Llama 4 Maverick-whose responses are independently assessed by two LLM judges. Together, the construction pipeline and evaluation framework provide a curriculum-grounded protocol for Vietnamese Informatics AI tutoring.**

***Index Terms*---AI tutoring, benchmark construction, pedagogical evaluation, large language models, Informatics education, Vietnamese education**

I. Introduction

Large language models (LLMs) produce increasingly correct and fluent answers, but effective tutoring also requires *(i)* interpreting the learner's understanding; *(ii)* diagnosing errors when evidence permits; *(iii)* selecting a pedagogical move; *(iv)* calibrating support; and *(v)* preserving the learner's opportunity to think. MathTutorBench shows that problem-solving ability does not directly translate into pedagogical quality [1]; KMP-Bench separates verifiable skills from principle-guided dialogue quality [2]; and TutorBench uses fine-grained, sample-specific criteria [3]. Tutor evaluation is therefore an assessment-design problem, not only a question-answering problem.

Prior AI-tutoring benchmarks leave three gaps for our setting. First, they concentrate on mathematics or broad STEM curricula rather than specialized school subjects. Second, they operate predominantly in English, leaving low-resource languages underserved. Third, construction method is not unified: KMP-Bench builds planned dialogue flows; TutorBench uses expert-authored situations and rubrics; and MathTutorBench organizes existing tutoring data under predefined tasks and metrics. These approaches structure data through prior task or situation assumptions. Less is known about converting ordinary teacher-authored pedagogical dialogues, collected independently of a benchmark taxonomy, into traceable evaluation units.

We address these gaps by constructing VEduTutorBench, a Vietnamese secondary-school Informatics AI-tutor benchmark for Grades 6--9. Rather than generating conversations from predefined benchmark situations, our construction and evaluation process:

- audits the structural, curricular, and pedagogical quality of raw teacher-authored dialogues;
- converts eligible tutor turns into traceable, context-conditioned next-response candidates;
- scores all six pedagogical principles, then applies deterministic filtering and locking; and
- combines Allison and Tharby's principles [4] with six tutor capabilities to construct context-specific rubrics and an AI-tutor evaluation protocol.

The paper makes three contributions:

1) an automated, traceable pipeline from raw dialogues to audited, principle-assigned, filtered, and locked benchmark samples;
2) a context-specific rubric library and evaluation method combining common and required-principle criteria; and
3) VEduTutorBench, with 1,400 completed Vietnamese Grades 6--9 Informatics tutoring samples across 665 raw dialogues.

II. Related Work and Background

A. Pedagogical and Measurement Foundations

Allison and Tharby's six principles describe complementary teaching functions rather than mutually exclusive task classes [4]. Adaptive-scaffolding research further distinguishes pedagogical moves from contingent support for a learner's current competence [5]. Evidence-centered design and contemporary validity theory motivate linking intended tutor capabilities to observable evidence and expert review [6], [7]. Accordingly, we use pedagogical principles to specify response-level requirements and six tutor capabilities to organize observable rubric criteria.

B. General AI-Tutoring Benchmarks

Existing benchmarks show that correctness alone does not capture tutoring quality. MathTutorBench distinguishes subject expertise, learner understanding, and pedagogical response generation [1]; KMP-Bench uses dialogue context and principle-guided criteria [2]; and TutorBench uses fine-grained, sample-specific criteria [3]. They motivate context-conditioned evaluation and explicit pedagogical requirements, but do not address our curriculum and data-construction setting.

C. Vietnamese and Informatics-Specific Educational Evaluation

Vietnamese evaluation resources establish language- and curriculum-aware assessment, but primarily evaluate examination answers, reasoning, or general conversation rather than context-conditioned tutor responses [8]--[10]. Vietnamese tutoring work has examined K--12 science questions, while Informatics-focused studies assess tutoring in block programming, robotics, or undergraduate programming contexts [11]--[14]. Thus, Vietnamese educational evaluation and Informatics-specific tutor evaluation have been addressed separately. Beside them, our literature search found no publicly documented benchmark combining Vietnamese Grades 6--9 Informatics, conversation-conditioned next-response evaluation, and traceable candidates derived from teacher-authored pedagogical dialogues.

III. Benchmark Construction Pipeline

Figure 1 summarizes the three-phase pipeline. Phase 1 audits raw teacher-authored dialogues; Phase 2 establishes six response-level principles and six broader tutor capabilities; and Phase 3 converts, labels, filters, and collects the benchmark samples.

A. Phase 1: Teacher-Authored Dialogues Audit

At Hanoi Metropolitan University (HNMU), teachers produced the 1,050 raw dialogues through an AI-assisted workflow grounded in Zone of Proximal Development (ZPD) and contingent scaffolding, which respectively frame assisted performance and the adaptive fading of support [15], [16]. Teachers defined exemplars and domain constraints, used LLM to draft by grade, and independently cross-reviewed and manually revised the outputs, shown detailed in (Appendix A-A). Phase 1 then audited these Grades 6--9 Vietnamese Informatics dialogues against eight curriculum sources, in 2 types: student textbooks (SGK) and teacher guides (SGV). After converting the source document into 154 OCR-derived SGK/SGV lesson-based Markdown units, the process normalized these units into 2,750 traceable fragments and a rebuildable SQLite full-text search index, used for the audit on each dialogue.

The audit followed a three-stage hybrid workflow. First, deterministic code normalized and traced the raw records, measured batch-level coverage, detected missing fields and formatting anomalies and identified exact or near-duplicate pairs. Second, the dialogues were reviewed by `hnmu-dialogue-auditor` AI agent, using `gpt-5.4-mini` at medium reasoning effort. It applied an 18-criterion checklist, which is validated and confirmed by HNMU's teachers, on raw fields of dialogue samples, SGK/SGV evidence retrieved from the full-text search index to evaluate each raw dialogue. There are 4 main criterion groups in the checklist: structure (3 criteria), consistency (7 criteria), pedagogy (6 criteria), and duplicate risk (2 criteria). Strict aggregation marked any failed criterion as *failed*, otherwise any uncertain criterion as *need human review*, and the rest as *passed*. The detailed checklist and examples of pass, uncertain, and fail cases are shown in Appendix A. At the end of this phase, 665 dialogues were retained, 382 were routed to human review due to uncertainty, and 3 were rejected.

TABLE I
Statistics of the 665 dialogues retained after Phase 1.

Metric	Value
Grade 6 dialogues	106 (15.94%)
Grade 7 dialogues	132 (19.85%)
Grade 8 dialogues	209 (31.43%)
Grade 9 dialogues	218 (32.78%)
Total dialogue turns	4,353
Student / tutor turns	2,325 / 2,028
Turns per dialogue (mean / median / range)	6.55 / 6 / 4--14
Characters per turn (mean / median / range)	86.5 / 79 / 7--294

B. Phase 2: Pedagogical Measurement Foundation

We distinguish response-level pedagogical *functions* from broader tutor *capabilities*. The functional layer adopts Allison and Tharby's Challenge, Explanation, Modelling, Practice, Feedback, and Questioning principles [4], operationalized for tutor evaluation by KMP-Bench [2]. They cover cognitive demand, clarification, demonstration, independent practice, learner-grounded response, and questions that advance the interaction.

The six-capability model synthesizes subject expertise, learner understanding, response generation, principle-guided behavior, adaptive explanation, feedback, and active learning from the three tutoring benchmarks [1]--[3]. Adaptive-scaffolding research [5] and HNMU guidance separate strategy selection from support calibration and treat diagnosis as input to contingent assistance ; evidence-centered design links each capability to observable response evidence [6].

The resulting capabilities are *(i)* subject-matter accuracy and grounding; *(ii)* learner-state modelling; *(iii)* pedagogical strategy selection; *(iv)* scaffolding calibration; *(v)* error, misconception, and prerequisite diagnosis; and *(vi)* clear, supportive, age-appropriate communication. Boundary analysis separates state description from causal diagnosis, and strategy selection from the amount, timing, and transfer of support. Principles activate response-specific criteria; capabilities organize their coverage.

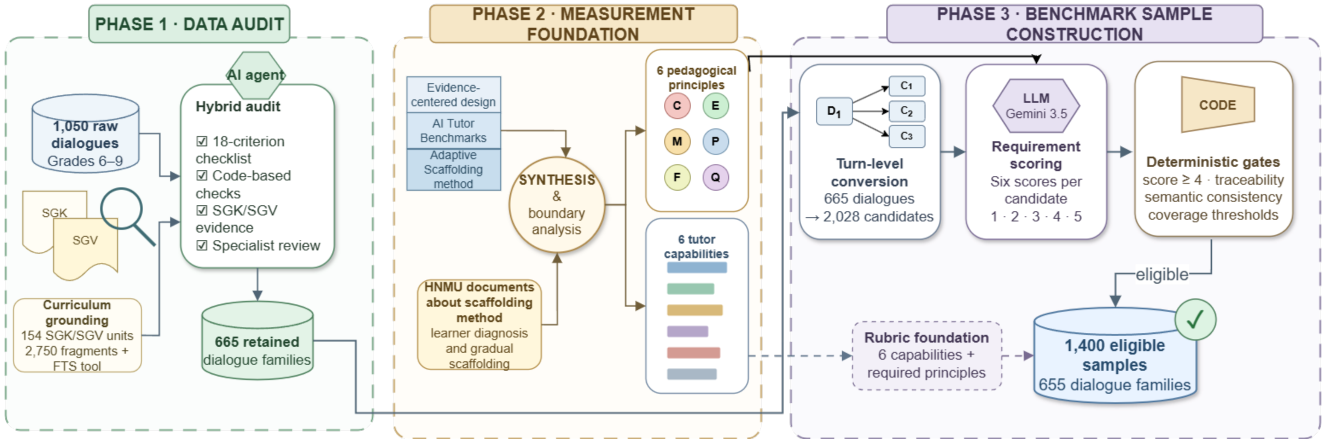

Fig. 1. Three-phase construction pipeline for VEduTutorBench.

C. Phase 3: From Raw Dialogues to Benchmark Data

Phase 3 combines audited dialogues and the pedagogical foundation through deterministic turn-level conversion, grounded principle-requirement scoring with code-based gates, and collection of the eligible pool to build the benchmark.

1) Turn-level candidate construction: Each tutor turn in the 665 retained dialogues becomes a possible reference response. The initial student turn is the prompt, intervening utterances retain their roles as history, and later turns are excluded; the source question and teacher-provided answer remain separate grounding. This deterministic policy produces 2,028 candidates across 665 families and multiple history depths. For example, a six-turn dialogue $D_1 = [S_1, T_1, S_2, T_2, S_3, T_3]$ yields 3 candidates: $C_1 = [S_1; \varnothing; T_1]$, $C_2 = [S_1; (T_1, S_2); T_2]$, and $C_3 = [S_1; (T_1, S_2, T_2, S_3); T_3]$, where each bracket records the prompt, history, and reference response, respectively. Here, S_i and T_i denote student and tutor turns. All three candidates retain the same family ID for tracing and data splitting.

2) LLM-assisted principle scoring and filtering: Gemini 3.5 Flash is used to score the necessity of all six principles from information available before the target response: grade, lesson, target position, Bloom level, student prompt, role-preserving history, source question, and teacher-provided answer. Excluding the reference response prevents criterion selection from being conditioned on the response used for comparison.

Scores range from 1 (irrelevant or counterproductive), through 2 (weak) and 3 (valid but optional), to 4 (required) and 5 (essential). Code selects principles scoring at least 4 and applies schema validation, thresholding, semantic linting, and eligibility gates. The executed Vietnamese prompt, its exact input structure, and a reader-facing English translation are provided in Appendix C.

3) Eligible benchmark pool and descriptive statistics: The run contains 12,168 scores for 2,028 candidates. Eligibility requires one to three principles; context-traceable evidence; an independent need and counterfactual for every score of 4 or 5; no optionality conflict (e.g. Feedback must guide improvement beyond praise, and Questioning must make an answer necessary rather than merely useful.); and a principle set appearing in at least five candidates from three families.

These gates retain 1,400 candidates from 665 Phase 1 dialogues and exclude 628. Overlapping flags comprise 592 confirmation-only Feedback cases, 20 inconsistent score-4/5 rationales, 10 Questioning cases without answer dependency, 8 cases with no score at least 4, 7 unsupported combinations, and one case requiring over three principles. Figure 2 summarizes principle, grade, turn, and length distributions.

IV. Evaluation Framework

Following KMP-Dialogue [2], the framework separates tutor input, criterion construction, and blind pairwise assessment.

A. Context Loading and Tutor-Response Generation

The tutor receives its Vietnamese Informatics role, grade, lesson, source task, common instruction, and required principles. Native multi-turn input preserves student turns as `user` and prior tutor turns as `assistant/model`; the final message is the student turn to answer. Sample IDs, scores, criteria, teacher-provided answer, and reference response remain hidden. Appendix D show the components used for a tutor response run, including their Vietnamese sample-specific system prompt format, native-role messages, and reader-facing translations.

B. Two-Tier Rubric Construction

The six capabilities map to four common criteria: *(i)* accuracy and verifiability; *(ii)* learner-state and goal alignment; *(iii)* calibrated support preserving learner work; and *(iv)* clear, respectful, age-appropriate communication. Each principle adds three criteria for its incremental function. The resulting 4 common and 18 principle-specific criteria received preliminary confirmation from HNMU Informatics teachers; a sample requiring n principles uses $4+3n$ criteria. Appendix B reports all six operational principles and the complete 22-rubric library.

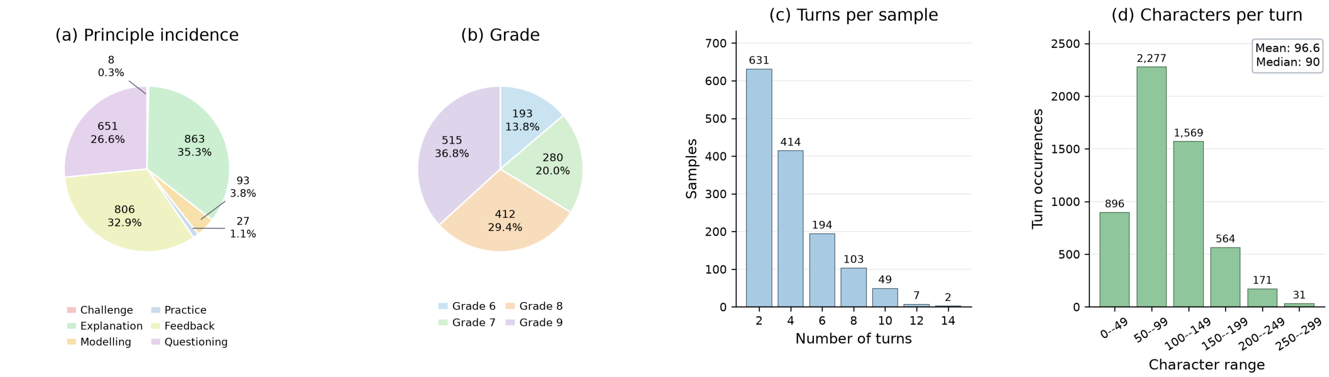

Fig. 2. Distributions of 1,400 eligible samples: (a) 2,448 principle assignments; (b) grade; (c) turns per sample, including prompt, history, and reference; and (d) characters per turn across 5,508 turns (mean 96.6; median 90).

C. Blind Pairwise Judging and Score Aggregation

The judge receives context, source question, teacher-provided answer, applicable criteria, and anonymous tutor generated and teacher-authored responses. Their order is deterministically randomized and restored after the judge selects *response 1*, *response 2*, or *Tie* per criterion. Code maps this to target-model Win, Lose, or Tie. For criterion r, with counts W_r, T_r, L_r, the win rate is

$$A_r = \frac{W_r}{W_r + T_r + L_r}.$$

The Vietnamese gold-answer-only judge prompt, deterministic user-message structure, and reader-facing translation are reported in Appendix E.

V. Experiments and Analysis

A. Experimental Setup

Metrics: Two LLM judges independently compare each tutor response with the teacher-authored reference: Gemini 3.5 Flash (`thinking_level=MEDIUM`) and `gpt-5.4-mini-2026-03-17` (medium reasoning). They share the prompt, 8,192-token cap, rubric set, anonymous ordering, and post-processing. Metrics are *(i)* holistic Overall Judgement Accuracy; *(ii)* General-Level Accuracy over four common criteria; *(iii)* Principle-Level Accuracy over three criteria per required principle; and *(iv)* Overall Accuracy, averaging General and mean Principle scores. Family-macro Overall weights each of 665 dialogue families equally. CIs and differences use 5,000 cluster-bootstrap resamples over `sample_id`; 4,995 cover all six principles.

Models: The panel comprises Gemini 3.5 Flash with baseline (baseline for short) and LearnLM-oriented (LearnLM for short) instructions, plus Llama 4 Maverick (Llama for short); the Gemini tutors are two instruction configurations of one base model. Both use `thinking_level=MEDIUM` without returned thoughts. All tutors omit sampling overrides, use a 2,048-token cap, and generate one response per sample: 4,200 responses for 1,400 samples.

B. Main Results

Table II shows both judges placing the Gemini configurations above Llama on General, Overall, and Family scores, with large recurring Questioning, Feedback, and Modelling gaps. Judge severity differs: Llama Overall is 49.51 under Gemini and 74.56 under GPT; Explanation is 62.96 versus 90.23, while Challenge is 87.50 under both but has only eight samples. The nearby Gemini configurations reverse order: Gemini favors baseline (87.87 vs. 85.35), whereas GPT slightly favors LearnLM (84.78 vs. 84.01). Thus, fine ordering and absolute values are judge-dependent, but the Gemini--Llama ordering is shared.

1) Pairwise Separation and Family-Level Robustness: Baseline, LearnLM, and Llama Overall CIs are respectively [86.46, 89.19], [83.76, 86.83], and [46.54, 52.37] under Gemini, and [82.36, 85.51], [83.28, 86.14], and [71.62, 76.97] under GPT. Table III shows all four Gemini--Llama difference intervals excluding zero (smallest lower bound: 7.43 points); only Gemini finds a non-zero baseline--LearnLM difference.

The holistic counts in Table IV reinforce the difference in judge severity toward Llama: Gemini records 603 holistic losses for the tutpr response, compared with 247 under GPT. Holistic judgment is reported separately and is not reconstructed from criterion-level votes.

C. Ablation Study

1) Instruction Ablation: Because the Gemini configurations differ only in instruction, Table V isolates this change. Gemini finds LearnLM reductions of 3.45 General, 4.25 Questioning, and 2.52 Overall points, with CIs excluding zero; all GPT and both holistic intervals include zero. Thus, no cross-judge evidence supports a LearnLM improvement, and the null Challenge result is weak with only eight samples.

2) Judge Robustness: Table VI reports exact agreement and Gwet's $AC1$, together with directional disagreements. Agreement is high for both Gemini configurations but falls to 63.21% for Llama. Llama disagreements are strongly asymmetric: the Gemini judge assigns Lose while GPT assigns Win in 419 cases, whereas the reverse occurs in only 65.

TABLE II

KMP-COMPATIBLE PAIRWISE WIN RATES BY JUDGE AND TUTOR CONFIGURATION (%).

Judge	Tutor Config.	OJ	GL	Challenge	Explanation	Modelling	Practice	Questioning	Feedback	OA	FM
Gemini	Gemini baseline	**93.43**	**83.27**	**100.00**	**97.88**	**98.92**	**91.36**	**76.65**	**90.03**	**87.87**	**88.49**
Gemini	Gemini+LearnLM	93.21	79.82	**100.00**	97.10	**98.92**	87.65	72.40	89.16	85.35	85.91
Gemini	Llama 4 Maverick	55.21	43.07	87.50	62.96	54.48	41.98	35.84	52.94	49.51	49.40
GPT	Gemini baseline	90.79	78.32	95.83	94.82	**98.21**	**91.36**	66.97	90.98	84.01	84.72
GPT	Gemini+LearnLM	**91.86**	**79.09**	**100.00**	**95.44**	**98.21**	90.12	**67.84**	**91.19**	**84.78**	**85.32**
GPT	Llama 4 Maverick	81.86	72.43	87.50	90.23	75.27	87.65	41.42	78.04	74.56	75.98

OJ: Overall Judgement; GL: General level; the six middle columns: principle level; OA: Overall; FM: family-macro OA. All values are pairwise win rates. Samples per configuration: Challenge 8, Explanation 863, Modelling 93, Practice 27, Questioning 651, and Feedback 806.

TABLE III

OVERALL-SCORE DIFFERENCES WITH 95% CLUSTER-BOOTSTRAP CIS.

Judge	Contrast	Difference	95% CI
Gemini	Baseline--LearnLM	+2.52	[1.53, 3.57]
Gemini	Baseline--Llama	+38.36	[35.53, 41.21]
Gemini	LearnLM--Llama	+35.84	[32.89, 38.82]
GPT	Baseline--LearnLM	-0.77	[-2.14, 0.45]
GPT	Baseline--Llama	+9.45	[7.43, 11.71]
GPT	LearnLM--Llama	+10.22	[8.04, 13.04]

TABLE IV

HOLISTIC WIN/TIE/LOSE RESULTS.

Judge	Tutor Config.	Win	Tie	Lose
Gemini	Gemini baseline	1,308 (93.43)	6 (0.43)	86 (6.14)
Gemini	Gemini+LearnLM	1,305 (93.21)	7 (0.50)	88 (6.29)
Gemini	Llama 4 Maverick	773 (55.21)	24 (1.71)	603 (43.07)
GPT	Gemini baseline	1,271 (90.79)	5 (0.36)	124 (8.86)
GPT	Gemini+LearnLM	1,286 (91.86)	9 (0.64)	105 (7.50)
GPT	Llama 4 Maverick	1,146 (81.86)	7 (0.50)	247 (17.64)

Parentheses show percentages; each row contains 1,400 comparisons.

TABLE V

PAIRED INSTRUCTION ABLATION ON THE SAME GEMINI BASE MODEL (%).

Component	N	Baseline	LearnLM	Δ [95% CI]
Gemini judge				
Overall Judgement	1,400	93.43	93.21	-0.21 [-1.65, 1.20]
General	1,400	83.27	79.82	-3.45 [-4.82, -2.06]
Challenge	8	100.00	100.00	0.00 [0.00, 0.00]
Explanation	863	97.88	97.10	-0.77 [-1.77, 0.19]
Modelling	93	98.92	98.92	0.00 [-1.05, 1.03]
Practice	27	91.36	87.65	-3.70 [-10.00, 0.00]
Questioning	651	76.65	72.40	-4.25 [-6.69, -1.77]
Feedback	806	90.03	89.16	-0.87 [-2.63, 0.99]
Overall Accuracy	1,400	87.87	85.35	-2.52 [-3.58, -1.50]
GPT judge				
Overall Judgement	1,400	90.79	91.86	+1.07 [-0.44, 2.64]
General	1,400	78.32	79.09	+0.77 [-0.75, 2.29]
Challenge	8	95.83	100.00	+4.17 [0.00, 14.29]
Explanation	863	94.82	95.44	+0.62 [-0.41, 1.66]
Modelling	93	98.21	98.21	0.00 [-1.06, 1.01]
Practice	27	91.36	90.12	-1.23 [-7.69, 3.57]
Questioning	651	66.97	67.84	+0.87 [-1.67, 3.44]
Feedback	806	90.98	91.19	+0.21 [-1.38, 1.75]
Overall Accuracy	1,400	84.01	84.78	+0.77 [-0.48, 2.14]

$\Delta =$ LearnLM minus baseline; CIs use 5,000 cluster-bootstrap draws over 665 families (4,999 for Challenge and Overall Accuracy because one draw has no Challenge sample).

TABLE VI

HOLISTIC CROSS-JUDGE AGREEMENT AND DISAGREEMENTS.

Scope	N	Exact	$AC1$	G1-L/G2-W	G1-W/G2-L	Tie
All configurations	4,200	80.45	0.775	--	--	--
Gemini baseline	1,400	88.36	0.874	57	95	11
Gemini+LearnLM	1,400	89.79	0.890	54	73	16
Llama 4 Maverick	1,400	63.21	0.529	419	65	31

Exact is a percentage. G1-L/G2-W and G1-W/G2-L denote opposite Gemini/GPT Lose--Win judgments; Tie means exactly one judge assigns Tie.

This pattern is consistent with different judge severity and compatible with a same-family effect, but does not identify the cause. The full directional matrix is retained in the machine-readable analysis bundle.

Across all 38,832 paired criterion decisions, exact agreement is 73.24%. Table VII shows substantial heterogeneity: Explanation and Modelling exceed 84% exact agreement, whereas Questioning reaches only 68.12%. The common criteria collectively reach 67.33%; within that group, appropriate communication is lowest at 61.90%. Challenge has the highest agreement but only 72 paired decisions. These results identify Questioning and the common criteria as priorities for rubric calibration; they do not by themselves reveal whether disagreement originates from the rubric wording, insufficient grounding, or judge behavior.

VI. DISCUSSION

A. *What the Evaluation Distinguishes*

The current benchmark robustly distinguishes relatively large differences in tutor-response quality. Both judges, sample-level and family-macro aggregation, and all four clustered Gemini--Llama comparisons consistently place the two Gemini configurations above Llama; the separation therefore does not depend on a single judge or on families contributing more samples.

Discrimination is less stable for similarly performing tutors. The Gemini baseline--LearnLM ordering reverses across judges, and only the Gemini judge finds a non-zero difference. The benchmark thus supports coarse separation of substantial differences, but not yet fine-grained ranking of close tutors.

B. *Measurement Implications*

The 22-rubric catalog---four common and 18 principle-specific criteria---received preliminary confirmation from HNMU Informatics teachers as appropriate to the target tutoring context. This supports its content relevance, although experts have not yet independently applied it to the evaluated responses. Explanation and Modelling show comparatively high cross-judge agreement, whereas Questioning and the common criteria require further calibration. Challenge and Practice are too scarce for strong principle-level claims. Because the KMP-compatible Overall score equally weights all six principles, Challenge with eight samples receives the same principle-level weight as Explanation with 863. This score supports

TABLE VII

Criterion-level agreement between judges.

Rubric group or criterion	N	Exact (%)	Gwet's $AC1$
All criteria	38,832	73.24	0.668
Common	16,800	67.33	0.583
Subject-matter accuracy	4,200	65.55	0.546
Learner-state and goal alignment	4,200	72.90	0.671
Calibrated support and agency	4,200	68.98	0.613
Appropriate communication	4,200	61.90	0.504 N
Challenge	72	95.83	0.956
Explanation	7,767	84.69	0.831
Modelling	837	86.74	0.850
Practice	243	77.78	0.736
Questioning	5,859	68.12	0.558
Feedback	7,254	76.88	0.726

denotes paired criterion decisions.

methodological comparability [2], but should be accompanied by principle support and family-sensitive results.

C. Threats to Validity and Next Validation

First, both judges are LLMs; their agreement measures cross-judge stability, not correctness against an independent human reference. HNMU expert teachers should next apply the same 22 rubrics independently to a stratified set of response pairs, then adjudicate disagreements into a human reference for measuring judge-model error and resolving Gemini--GPT conflicts. This stage should prioritize the common accuracy, communication, and scaffolding criteria, Questioning, and directional judge disagreements.

Second, the Gemini judge belongs to the same model family as two evaluated tutors. This may reflect a same-family preference or merely different judge severity. The current scores cannot distinguish them: Gemini reports Gemini--Llama Overall gaps of 35.84--38.36 percentage points, versus 9.45--10.22 for GPT. Larger Gemini-judge gaps therefore should not be treated as evidence of same-family bias without the human reference above.

Finally, rubric boundaries and model-assigned required-principle sets require broader calibration. Principles were activated by one Gemini scoring run at `requirement_score >= 4`. Further validation should test rubric boundaries without editing them merely to maximize judge agreement, review the activated principles, and add Challenge, Practice, and Modelling cases before fine-grained principle-level claims.

VII. Conclusion

We introduced VEduTutorBench, a Vietnamese Grades 6--9 Informatics tutoring benchmark constructed from teacher-authored dialogues through auditable screening, turn-level conversion, grounded pedagogical-principle requirement scoring, and deterministic eligibility gates. Its 1,400 samples support native multi-turn tutor generation and context-specific evaluation with four common criteria plus three criteria per required principle. Across three tutor configurations and two separately reported LLM judges, the framework consistently distinguishes the larger Gemini--Llama performance gap, while the Gemini baseline--LearnLM ordering remains judge-dependent. These results support relative discrimination of large differences, not absolute quality certification; expert calibration and controlled model-bias testing remain necessary next steps for future work.

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Appendix A
Phase 1 Agent-Audit Checklist and Examples

A. HNMU Raw-Dialogue Authoring Workflow

HNMU supplied an internal theoretical training deck, a scaffolding framework with illustrative dialogues, and an operational authoring protocol. The materials connect ZPD with contingent scaffolding: support should bridge what a learner can do independently and with guidance, become more specific after difficulty, fade after success, and progressively return responsibility to the learner [15], [16]. HNMU documented the following teacher-led workflow:

1) **Establish the pedagogical foundation.** Teachers studied ZPD and contingent scaffolding as the primary theoretical anchors; the training materials also used cognitive apprenticeship and Socratic questioning as complementary design ideas.
2) **Define curriculum grounding.** Teachers selected grade-specific Bloom expectations and official Vietnamese Informatics textbooks, teacher guides, workbooks, and curriculum requirements as the permitted knowledge base.
3) **Specify quality criteria.** The expert team established the 18 criteria covering structure, content consistency, pedagogy and scaffolding, and duplicate risk.
4) **Construct exemplars.** Teachers selected textbook exercises and, based on them, wrote tutoring interactions sequence intended to satisfy all 18 criteria.
5) **Design constrained prompts.** Teachers encoded the criteria, exemplars, and curriculum boundaries into prompts for dialogue generation.
6) **Generate grade-specific draft diaglogues.** LLM produced dialogue drafts, with designated teachers responsible for individual grade groups.
7) **Review, cross-check, and revise.** An independent expert reviewed each generated dialogue against the criteria; teachers from other grade groups then cross-checked curricular continuity and overload or repetition risks. Flagged dialogues were manually edited before inclusion in the 1,050-dialogue raw collection.

Thus, *teacher-authored* denotes teacher ownership of the theoretical framework, criteria, exemplars, prompts, review decisions, and final revisions; LLM supported draft production rather than final acceptance.

B. Agent-Audit Checklist

The subsequent Phase 1 workflow also contained deterministic batch-level and process checks. This appendix reports the 18 criteria applied to every dialogue by the `hnmu-dialogue-auditor`. The agent evaluated each criterion independently as *pass*, *fail*, *uncertain*, or *not applicable*. It received the raw fields, mechanical flags produced by code, retrieved SGK/SGV evidence, and the HNMU guidance on scaffolding and cognitive levels. Table VIII gives the operational checks.

TABLE VIII: The 18 criteria in the per-dialogue agent audit.

ID	Group	Criterion	Operational check and evidence
`RAW-STR-02`	Structure	Core fields are present	Check that the question, teacher-guide answer, and dialogue contain enough information to be audited; use code-generated missing-field flags as input.
`RAW-STR-03`	Structure	Speaker labels are valid	Check that learner and tutor turns can be parsed reliably; use mechanical speaker-label flags and the raw dialogue.
`RAW-STR-04`	Structure	Dialogue is sufficiently complete	Check that the dialogue has enough turns and substantive content for a semantic and pedagogical audit; use code-generated length flags as supporting signals.
`RAW-CON-01`	Consistency	Question matches lesson and position	Compare the question and recorded location against retrieved SGK/SGV content from the same grade and lesson.
`RAW-CON-02`	Consistency	Teacher-guide answer matches the question	Determine whether the recorded SGV answer correctly and sufficiently answers the question, using SGV evidence when available.
`RAW-CON-03`	Consistency	Dialogue addresses the question	Determine whether the interaction responds to the learner's stated problem rather than drifting to a different task.
`RAW-CON-04`	Consistency	Dialogue is consistent with the answer	Determine whether the tutor guides the learner toward content that is consistent with the teacher-provided answer and SGV evidence.
`RAW-CON-05`	Consistency	Cognitive level is appropriate	Compare the recorded Bloom level with the action, object, and demand of the question under the normalized HNMU cognitive-level guidance.
`RAW-CON-06`	Consistency	No unsupported curriculum content	Use retrieved SGK/SGV evidence to detect claims that are fabricated or contradict the relevant curriculum content.
`RAW-CON-07`	Consistency	Metadata and content are mutually consistent	Check whether grade, lesson, position, question, answer, and dialogue refer to the same curricular content, using the lesson map and retrieved evidence.
`RAW-PED-01`	Pedagogy	Scaffolding is present	Check for elicitation, hints, staged support, or understanding checks before the tutor concludes.
`RAW-PED-02`	Pedagogy	The answer is not revealed prematurely	Check whether the tutor gives the complete answer or solution before diagnosis, learner effort, or an appropriate level of support.
`RAW-PED-03`	Pedagogy	The dialogue sequence is coherent	Check whether successive turns respond to the learner's evolving state and form a coherent progression from support to consolidation.
`RAW-PED-04`	Pedagogy	Turns have learning value	Identify redundant turns, generic praise, or repetition that adds no useful feedback, hint, explanation, or next step.
`RAW-PED-05`	Pedagogy	Language is age appropriate	Check whether wording, length, and tone are suitable for Grades 6--9 learners; route ambiguous cases to HNMU review.

Continued on next page

TABLE VIII continued

ID	Group	Criterion	Operational check and evidence
RAW-PED-06	Pedagogy	No off-task substitution	Check whether the tutor avoids or replaces the learner's task with unrelated content or generic advice.
RAW-DUP-03	Duplicate risk	Not a trivial variant	Use code-generated duplicate candidates and the sample texts to decide whether a dialogue differs from another only by minor words, values, or entities.
RAW-DUP-04	Duplicate risk	No mechanically repeated AI template	Use dialogue patterns in the assigned shard or batch to identify repetitive praise, prompting, or scaffolding structures that materially reduce diversity.

C. Audited Examples

For each example, the original Vietnamese record is presented first. The English text is a reader-facing translation and was not used by the audit pipeline. Internal sample and learning-resource identifiers are omitted.

1) Passed Example: **Original Vietnamese record (used in the audit)**

Lớp: 9
Bài học: [13b]: Biên tập và xuất video
Câu hỏi: Phân biệt được tệp dự án (Project) và tệp video đã xuất (Exported video).
Đáp án từ sách giáo viên: Tệp dự án chứa các lệnh chỉnh sửa, chỉ mở bằng phần mềm làm video. Tệp video đã xuất (.mp4) là thành phẩm, có thể xem bằng mọi trình phát video.
Hội thoại:
HS: Em lỡ gửi cái tệp Backup (.vbp) của bài trước cho bạn em xem, bạn bảo không xem được hình gì cả.
AI: Em còn nhớ tệp Backup (.vbp) đó chứa cái gì không?
HS: Nó chứa dữ liệu hình ảnh, âm thanh để mở lại phần mềm làm tiếp ạ.
AI: Đúng. Để xem được nó, bạn em cũng phải mở phần mềm Video Editor lên. Nhưng bạn em muốn "xem phim" như trên YouTube, thì em phải gửi định dạng nào?
HS: Em phải gửi tệp phim đã được xuất ra, là định dạng .mp4 ạ?
AI: Chuẩn xác. Tệp dự án là nguyên liệu để nấu ăn, còn tệp mp4 là món ăn đã nấu chín. Khán giả chỉ cần món ăn thôi.
HS: Dạ em hiểu rõ sự khác biệt rồi. Em sẽ vào phần mềm xuất lại ra mp4 gửi bạn.
AI: Rất tốt, sự cố đã giúp em ghi nhớ sâu sắc bản chất của tệp.

Reader-facing English translation (not used for inference)

Context: Grade 9, Lesson 13B, Editing and Exporting Video.
Question: Distinguish a project file from an exported video file.
Teacher-guide answer: A project file stores editing instructions and is reopened in the video editor; an exported .mp4 file is the finished product and can be viewed in media players.
Dialogue:
STUDENT: I accidentally sent my friend the Backup file (.vbp) from my previous project, but they said they could not see anything.
TUTOR: Do you remember what that Backup file contains?
STUDENT: It contains the image and audio data so that I can reopen the project in the software and continue editing.
TUTOR: Correct. To view it, your friend would also need to open the video editor. But if your friend wants to watch the film as on YouTube, which format should you send?
STUDENT: I should send the exported film file, in .mp4 format?
TUTOR: Exactly. A project file is like the ingredients for cooking, whereas the .mp4 file is the finished dish. The audience only needs the finished dish.
STUDENT: I understand the difference now. I will reopen the software, export the video to .mp4, and send that file.
TUTOR: Very good. This incident has helped you understand the nature of the two file types more deeply.

All 18 criteria passed. The tutor elicited the learner's current understanding, prompted an inference about the appropriate output format, and consolidated the distinction with an analogy. Strict aggregation retained the dialogue as passed.

2) Uncertain Example: **Original Vietnamese record (used in the audit)**

Lớp: 8
Bài học: [11b] Thực hành tổng hợp
Câu hỏi: Giải thích được sự khác biệt giữa lưu tệp dự án (Save) và xuất tệp ảnh (Export As).
Đáp án từ sách giáo viên: Save (Ctrl+S): Lưu dự án gốc đuôi .xcf, giữ nguyên các layer để sau này mở ra sửa tiếp. Export As (Shift+Ctrl+E): Xuất ra tệp ảnh phẳng .jpg/.png để in ấn, gửi cho người khác xem.
Hội thoại:
HS: Em thiết kế xong cái Poster rồi. Sách giáo khoa bảo chọn "File / Export As để ghi lại ảnh với tên SchoolTour.jpg". Sao mình không ấn "Save" cho nhanh ạ?
AI: Một câu hỏi rất hay về quản lí file! Thầy giả sử em là kiến trúc sư. Lệnh "Save" (đuôi .xcf) giống như em lưu lại bản vẽ có dính đầy ghi chú, tẩy xoá, các bản nháp xếp chồng lên nhau. Còn "Export As" (đuôi .jpg) giống như em in ra một bức tranh phối cảnh hoàn mĩ để đưa cho khách hàng.
HS: Dạ, tức là Save là để mình giữ lại bản gốc sửa sau, còn Export là để nộp bài ạ?
AI: Em tóm tắt cực kì xuất sắc! Khi em Export ra .jpg, phần mềm sẽ "ép" tất cả các Layers lại thành một mặt phẳng duy nhất. Ưu điểm là dung lượng rất nhẹ và máy tính nào, điện thoại nào cũng xem được. Nhược điểm là gì em biết không?
HS: Là các lớp dính vào nhau hết, không lôi cái ảnh ra sửa hay đổi màu chữ được nữa ạ.
AI: Tuyệt vời. Chính vì vậy, quy trình chuẩn của dân chuyên nghiệp là: Nhấn Save (.xcf) để cất làm vốn trước, sau đó mới nhấn Export As (.jpg) để đem đi nộp. Em làm y như vậy nhé.
HS: Dạ vâng, em ghi nhớ quy tắc này rồi ạ.

Reader-facing English translation (not used for inference)

Context: Grade 8, Lesson 11B, Integrated Practice.
Question: Explain the difference between saving a project (Save) and exporting an image (Export As).
Teacher-guide answer: Save stores the editable .xcf project with its layers, whereas Export As produces a flattened .jpg or .png image for printing or sharing.
Dialogue:
STUDENT: I have finished designing my poster. The textbook says to choose File / Export As and save it as SchoolTour.jpg. Why can I not simply press Save?
TUTOR: Imagine that you are an architect. Save (.xcf) is like storing the original drawing together with its notes, erasures, and stacked drafts. Export As (.jpg) is like printing a finished rendering for a client.

STUDENT: So Save keeps the original for later editing, whereas Export produces the file that I submit?
TUTOR: Excellent summary. When you export to .jpg, the software flattens all layers into one image. The file is small and can be viewed on almost any computer or phone. What is the disadvantage?
STUDENT: The layers are merged, so I cannot separately move the picture or change the text colour afterwards.
TUTOR: Exactly. The standard workflow is to save the .xcf project first, and then export a .jpg file for submission. Please follow those steps.
STUDENT: Yes, I will remember that rule.

Seventeen criteria passed. RAW-PED-02 was uncertain because the first tutor response nearly completed the comparison before eliciting the learner's reasoning. Later turns re-engaged the learner and checked understanding, so strict aggregation routed the dialogue to review rather than rejecting it.

3) Failed Example: **Original Vietnamese record (used in the audit)**

Lớp: 7
Bài học: Bài 1. Thiết bị vào - ra
Câu hỏi: Chức năng của loa và tai nghe có điểm gì giống nhau?
Đáp án từ sách giáo viên: Đều là thiết bị ra, dùng để xuất thông tin dạng âm thanh từ máy tính.
Hội thoại:
HS: Loa với tai nghe thì quá rõ rồi, nhưng trong bài học chúng có chung đặc điểm gì ạ?
AI: Em hãy nhớ lại 4 thành phần xử lý thông tin. Cả hai thiết bị này đều đưa thứ gì đó đến tai em?
HS: Dạ đưa âm thanh ạ.
AI: Âm thanh đó đi từ "trong" máy ra hay từ ngoài vào máy?
HS: Dạ từ trong máy ra ạ.
AII: Vậy theo bảng phân loại, chúng cùng thuộc nhóm thiết bị gì nhỉ?
HS: Dạ là thiết bị ra! Em hiểu rồi.

Reader-facing English translation (not used for inference)

Context: Grade 7, Lesson 1, Input and Output Devices.
Question: What do speakers and headphones have in common in terms of function?
Teacher-guide answer: Both are output devices that convey audio information from a computer.
Dialogue:
STUDENT: Speakers and headphones are familiar devices, but what feature do they share in this lesson?
TUTOR: Recall the components involved in information processing. What do both devices deliver to your ears?
STUDENT: Sound.
TUTOR: Does that sound travel from inside the computer to the outside, or from outside into the computer?
STUDENT: From inside the computer to the outside.
AII: According to the classification table, which device group do they both belong to?
STUDENT: They are output devices. I understand now.

RAW-STR-03 failed because the raw dialogue used the invalid speaker label AII instead of AI. RAW-PED-02 and RAW-PED-03 were consequently uncertain because reliable turn parsing was compromised; the remaining 15 criteria passed. Strict aggregation classified the dialogue as failed, although its speaker label could be repaired.

Appendix B
Pedagogical Principles and Rubric Library

Tables IX and X report the complete operational constructs used by the benchmark. As in Table VIII, all table content is presented in English for readers. These are reader-facing translations of the canonical Vietnamese wording used by the prompts and judging workflow; the translations were not used for inference or evaluation. Internal workflow metadata is omitted. The 22-rubric library contains four common criteria and three principle-specific criteria for each of the six principles. Each row is scored pairwise: Win when the tutor response is better than the teacher-authored reference on that criterion, Tie when they are equivalent, and Lose when it is worse or fails the criterion. The final column provides criterion-specific comparative anchors.

TABLE IX: The six pedagogical principles and their operational definitions.

Principle	Definition	Include when	Exclude when	Observable response evidence
Challenge	Set or maintain a cognitive demand that is sufficiently difficult yet attainable, creating productive effort and helping the learner progress beyond current performance.	The response asks the learner to reason more deeply, handle a more difficult case, or overcome a deliberate obstacle independently.	Do not assign merely because the lesson is difficult, the tutor asks a question, or the response increases workload without fitting the learner state.	A clear cognitive demand is calibrated to the current level and leaves meaningful thinking to the learner.
Explanation	Make a concept, relationship, principle, method, or rationale clear and concrete in a way that fits the learner state.	The response must help the learner understand what, how, or why rather than merely provide an answer or an operational step.	Do not assign when the response only confirms, only asks a question, or only demonstrates a procedure without a substantive explanatory function.	The response clarifies the core idea, connects it to the context, and does not merely reproduce a definition.
Modelling	Let the learner observe how knowledge is applied through a process, reasoning path, decision path, operation, or model product.	The response demonstrates a model or part of one and makes the method of acting or thinking visible.	Do not assign to every illustrative example, isolated hint, or complete solution that lacks a transfer purpose.	A concrete and correct model exposes decision points and gives the learner an opportunity to continue independently.
Practice	Require the learner to perform or repeat an application of knowledge or skills to strengthen retention, fluency, or independent performance.	The response assigns or continues an activity intended to build fluency, test application, or consolidate recent learning.	Do not assign to a question that only gathers information, a required step in the current solution, or a new challenge that is not intended as practice.	The learner performs an activity aligned with the learning objective, at an appropriate difficulty, without receiving the complete solution immediately.
Feedback	Use an observed learner answer, method, operation, or product as the object of evidence-based comment that guides improvement.	The learner has produced an answer, product, operation, or line of reasoning that the tutor needs to evaluate.	Do not assign to social praise, unsupported confirmation, or an explanation that does not respond directly to learner work.	The response identifies observable details, explains their significance, and provides an appropriate correction or next step when needed.
Questioning	Require a learner answer in order to diagnose understanding, sustain reasoning, or promote deeper thinking.	A question serves a clear pedagogical function, and the learner answer will help the tutoring interaction progress.	Do not assign to social, rhetorical, vague, or immediately self-answered questions.	The question is contextually grounded, has a clear cognitive purpose and suitable openness, and genuinely requires a learner answer.

TABLE X: The complete 22-rubric library.

ID / scope	Criterion	Observable evidence	Boundary	Comparative anchors
`RUB-GEN-ACC` General	**Subject-matter accuracy and verifiability**	Concepts, relationships, examples, operations, code, procedures, and results are consistent with the source question, gold_answer, and traced textbook or teacher-guide evidence; no source is fabricated.	This criterion measures correctness and grounding only; friendliness, length, or pedagogical strategy cannot compensate for incorrect content.	**Positive:** Content is correct, states the decisive details, and makes no claim beyond the evidence. **Near miss:** Content is broadly correct but omits a useful detail without changing the conclusion or misleading the learner. **Negative:** A concept, procedure, code fragment, or source claim is wrong and may cause incorrect learning or action.
`RUB-GEN-ALIGN` General	**Alignment with learner state and immediate goal**	The response uses what the learner has asked, tried, understood, misunderstood, or currently needs and does not ignore decisive information in the history.	Describing the current learner state differs from explaining the root cause of an error; diagnosis is required only when an applicable principle-specific criterion and sufficient evidence support it.	**Positive:** The response directly addresses the current difficulty and goal and uses necessary information from the history. **Near miss:** It addresses the right topic but remains generic or overlooks an important detail about prior learner work. **Negative:** It answers a different question, repeats what the learner already knows, or assumes a state contradicted by the input.

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TABLE X continued

ID / scope	Criterion	Observable evidence	Boundary	Comparative anchors
RUB-GEN-SCAFF General	**Calibrated support preserving meaningful learner work**	The amount of information, number of steps, degree of hinting, and remaining learner work fit the current state; the response neither substitutes for a learner who can continue nor abandons a learner who is stuck.	This criterion measures the level and timing of support rather than whether each principle-specific behavior is present.	**Positive:** The response gives enough support for the learner to advance while preserving meaningful thinking. **Near miss:** The strategy is reasonable but gives slightly too much or too little support; progress remains possible, although agency is reduced or more help is needed. **Negative:** It reveals the entire solution prematurely or merely tells a demonstrably stuck learner to try again.
RUB-GEN-COMM General	**Clear, respectful, age-appropriate communication**	Language is accessible, structure avoids overload, and address is respectful; specific effort or progress is acknowledged when appropriate, without belittling, pressure, or generic praise replacing support.	Evaluate only communication behavior in the current response; do not infer learner motivation, emotions, or actual learning outcomes.	**Positive:** The response is concise, easy to follow, respectful, and acknowledges specific learner work before giving direction. **Near miss:** It is polite and understandable but somewhat long, dense, or generically encouraging. **Negative:** It is sarcastic, attributes errors to learner ability, causes embarrassment, uses inaccessible language, or is disorganized.
RUB-CHA-DEMAND Challenge	**Meaningful cognitive demand**	The response sets or maintains an obstacle, question, or task that requires reasoning, choosing, generalizing, or deeper application rather than mechanical recall.	Do not rate highly merely because a question is difficult, long, or adds work; the difficulty must serve the learning objective.	**Positive:** The learner must justify a method, compare alternatives, or handle a related new case. **Near miss:** Only values are changed or an almost mechanical extra step is added, creating work with little cognitive value. **Negative:** Difficulty is unrelated, trick-based, or far beyond the current content and goal.
RUB-CHA-CALIB Challenge	**Challenge calibrated to the current state**	Challenge and support fit prior learner performance, the current error, grade, and cognitive level and provide a viable starting point.	Unlike RUB-CHA-DEMAND, this criterion measures attainability rather than whether the task itself prompts deep thinking.	**Positive:** The challenge exceeds the just-completed level while using available knowledge and providing an entry hint when needed. **Near miss:** It is relevant but slightly too easy or difficult; a minor adjustment would make it feasible. **Negative:** It is too easy to require effort or too difficult and unsupported to be attainable.
RUB-CHA-AGENCY Challenge	**Core thinking remains with the learner**	The response states a target or constraints without solving the very reasoning that the challenge is meant to elicit; the learner must still produce a meaningful answer.	This is not equivalent to supplying little information; the tutor may provide data or hints that do not remove the target reasoning.	**Positive:** Necessary conditions are supplied, but the learner must predict, justify, or verify. **Near miss:** A challenge remains, but hints nearly determine the answer. **Negative:** The tutor solves the challenge or asks the learner only to repeat a supplied conclusion.
RUB-EXP-CORE Explanation	**Core idea and decisive relations are clarified**	The response explains what, how, or why by clarifying a concept, causal relation, rule, or error cause rather than merely stating an answer.	Naming, listing steps, or giving a correct answer does not count as explanation when the relation or rationale remains absent.	**Positive:** It explains why two quantities must use the same unit before division and what happens otherwise. **Near miss:** It states the correct rule, but the reason remains unclear or simply paraphrases the question. **Negative:** It only supplies a result, term, or operational command without improving understanding.
RUB-EXP-COHER Explanation	**Coherent and grounded explanatory flow**	Ideas follow an understandable relation, use a suitable example or contrast, and connect directly to the current question or error.	Unlike Modelling, an example here clarifies relationships; Modelling must demonstrate a procedure or decision path.	**Positive:** A short example connects the bit concept to two states and returns to the learner question. **Near miss:** Information is correct but disconnected, or the example is distant or weakly tied to the conclusion. **Negative:** The explanation is meandering, internally contradictory, or uses an example that distorts the core relation.
RUB-EXP-ADAPT Explanation	**Learner-appropriate detail and wording**	Depth, terminology, examples, and information volume fit the grade, cognitive level, and prior knowledge; the entire lesson is not repeated when only one link is missing.	This criterion does not re-evaluate politeness; it measures adaptation of the explanation itself.	**Positive:** It explains only the missing conversion step in Grade-appropriate language and leaves the learner to continue the calculation. **Near miss:** It is correct but slightly too long, too brief, or contains an unexplained term. **Negative:** It delivers a full lecture without considering learner state or uses abstractions the learner cannot follow.

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TABLE X continued

ID / scope	Criterion	Observable evidence	Boundary	Comparative anchors
RUB-MOD-PROC Modelling	**Observable demonstration of a process or method**	The response demonstrates part of an operation, algorithm, code sequence, product, or decision path in the correct order and sufficiently shows how to apply it.	Not every example or step list is modelling; a method must be demonstrated on a concrete case.	**Positive:** It demonstrates one binary-search split, records the left and right bounds, and updates the search interval for a concrete value. **Near miss:** It lists the correct steps but does not apply them to a case, making the process difficult to observe. **Negative:** It gives only the final answer, demonstrates an incorrect method, or completes the problem without exposing the process.
RUB-MOD-THINK Modelling	**Decision points and reasoning are made visible**	The response explains why an operation, condition, branch, or representation is selected at important points so that the learner can reuse the reasoning.	Private model chain-of-thought is not required; a brief pedagogical explanation of verifiable decisions is sufficient.	**Positive:** In a debugging model, it identifies the loop condition that must change and explains why the current value prevents termination. **Near miss:** The demonstration is correct, but reasons at decision points remain implicit. **Negative:** The response asks the learner to imitate clicks without explaining when or why to use them.
RUB-MOD-TRANSFER Modelling	**Modelling supports transfer rather than substitution**	The response limits what is modelled, identifies cues for continued application, and leaves an analogous part for the learner.	Unlike Practice, this criterion evaluates the design of the model; Practice requires an activity with a clear consolidation function.	**Positive:** It models one step and asks the learner to apply the same rule to the next step or case. **Near miss:** It models almost everything and leaves only a mechanical operation. **Negative:** It gives a complete solution without a transfer point or opportunity for learner application.
RUB-PRA-ACT Practice	**Learner-performed activity**	The response assigns a question, operation, exercise, or case requiring the learner to apply knowledge or skill and does not answer it immediately.	A required step in the current problem is not automatically Practice when its sole purpose is to finish that solution rather than create consolidation.	**Positive:** After an example is understood, the learner receives a similar case to perform or explain independently. **Near miss:** The learner is asked to act, but the action is too small or only repeats what was just stated. **Negative:** The tutor performs the activity, merely asks whether the learner understands, or creates no learner action.
RUB-PRA-ALIGN Practice	**Practice aligned with target skill and learner state**	The activity targets the knowledge or skill that needs consolidation, fits the grade and recent learning or error, and has a feasible, unambiguous answer.	Unlike Challenge, Practice prioritizes consolidation and independence and need not substantially increase cognitive difficulty.	**Positive:** After supported success, the learner receives a new case using the same skill of distinguishing input and output devices. **Near miss:** The activity matches the topic but not the specific error, or almost duplicates the prior example. **Negative:** It targets the wrong skill, contains faulty data, or is far too easy or difficult for the state.
RUB-PRA-CONSOL Practice	**Consolidation toward independent performance**	The response introduces enough purposeful variation or repetition for the learner to recall and apply the skill without immediately revealing the solution.	Multiple exercises are not required; one activity is sufficient when it clearly consolidates learning and preserves learner performance.	**Positive:** Context or data change while the target skill remains, and the learner performs independently before reporting the result. **Near miss:** The activity is useful but too similar to the model or too heavily hinted to require independence. **Negative:** The answer is supplied immediately, repetition is purposeless, or added workload does not support consolidation.
RUB-FBK-GROUND Feedback	**Feedback grounded in observed learner work**	The comment points to a specific learner answer, operation, line of reasoning, or product rather than offering only praise, criticism, or confirmation.	Root-cause diagnosis is not always required; when the input shows only an outcome, do not infer a deeper misconception.	**Positive:** It notes that the learner selected division correctly but that the two quantities still use different units. **Near miss:** It says the answer is almost correct without identifying what is right or what needs review. **Negative:** It offers generic praise, says the answer is wrong, or confirms it without reference to observable details.

Continued on next page

TABLE X continued

ID / scope	Criterion	Observable evidence	Boundary	Comparative anchors
RUB-FBK-DISC Feedback	**Correct and improvable elements are distinguished**	The response accurately identifies the quality of learner work, prioritizes the decisive error or progress, and briefly explains why it matters.	Unlike Explanation, this criterion begins with and evaluates learner work; a general explanation is not automatically Feedback.	**Positive:** It confirms the correct search direction, identifies an overly broad keyword, and explains why a school or location name is needed. **Near miss:** The comment is correct but misses an important issue or does not explain its effect on the result. **Negative:** It misjudges learner work, reinforces an error, or substitutes an answer without distinguishing what should change.
RUB-FBK-ACTION Feedback	**A feasible improvement step is provided**	When improvement remains, the response offers a specific, calibrated question, hint, or next step; after completion, it gives grounded confirmation and extends only with a clear purpose.	A new task is not always required, but praise or confirmation alone is insufficient when an unmet learning need remains.	**Positive:** It asks the learner to recheck the number of elements at the boundary instead of repairing the entire array. **Near miss:** It asks for another check but gives no region or starting method. **Negative:** It offers no improvement direction, gives incorrect direction, or performs the whole correction.
RUB-QUE-PURPOSE Questioning	**Clear pedagogical purpose**	The question diagnoses understanding, sustains reasoning, retrieves needed knowledge, or promotes deeper thinking, and the expected answer informs the next turn.	Social, rhetorical, decorative, or generic comprehension-check questions do not constitute high-quality Questioning.	**Positive:** It asks where the loop condition changes to distinguish an update error from a condition error. **Near miss:** The question is relevant, but its purpose is unclear or the answer would not change subsequent support. **Negative:** It is perfunctory, immediately self-answered, or unrelated to the current goal.
RUB-QUE-QUALITY Questioning	**Clear, answerable, appropriately demanding question**	The question is accurate, contextually grounded, sufficiently scoped, offers an entry point, uses grade-appropriate language, and has openness suited to the goal.	Unlike Challenge, a question may diagnose or retrieve knowledge without introducing greater difficulty.	**Positive:** It asks for a specific sign in code or the interface that can determine the next debugging step. **Near miss:** It is directionally correct but too broad, combines two demands, or contains an unexplained term. **Negative:** It is vague, underspecified, based on a false premise, or beyond the current level.
RUB-QUE-DEPEND Questioning	**The learner answer is genuinely required**	The response stops at an appropriate point for the learner to answer, neither immediately answers nor buries the question after a complete solution, and can use the answer to guide later support.	This criterion measures interaction that depends on an answer rather than question difficulty or content alone.	**Positive:** It asks a diagnostic question and waits for the learner to describe the symptom before suggesting a repair. **Near miss:** It places a question after nearly completing the solution, leaving little value in the learner answer. **Negative:** It asks and answers a rhetorical question or requests a formal confirmation after providing the entire answer.

Appendix C
Requirement-Scoring Prompt

This appendix reproduces the requirement-scoring prompt used for the full 2,028-candidate run. Vietnamese text below is the executed input; the English translation is provided only for readers and was not used for inference.

A. Executed Vietnamese System Prompt

```
Bạn là bộ chấm mức độ cần thiết của sáu nguyên tắc sư phạm đối với phản
hồi tiếp theo của gia sư AI môn Tin học THCS Việt Nam.

## Nhiệm vụ

Với đúng một JSON đầu vào, chấm độc lập cả sáu nguyên tắc bằng
`requirement_score` từ 1 đến 5. Câu hỏi duy nhất cần trả lời là:

> Để phản hồi tiếp theo đáp ứng đúng nhu cầu quan sát được của học sinh,
> nguyên tắc này cần thiết ở mức nào?

Bạn không chấm chất lượng của một phản hồi gia sư đã có. Bạn không được
chọn nguyên tắc chỉ vì nó có thể hữu ích hoặc vì bạn thích chiến lược đó.

## Ý nghĩa tám trường đầu vào

- `grade`: lớp 6–9; dùng để hiệu chỉnh mức kiến thức và độ phù hợp lứa
  tuổi, không dùng để suy đoán năng lực riêng của học sinh.
- `lesson`: bài học hoặc chủ đề Tin học của tình huống.
- `position`: mục hoặc trang trong học liệu nguồn, không phải vị trí lượt
  hội thoại.
- `bloom_level`: mức nhận thức dự kiến; không tự động làm tăng điểm
  `Challenge`.
- `student_prompt`: phát biểu hoặc yêu cầu mở đầu của học sinh.
- `conversation_history`: các lượt từ sau `student_prompt` đến ngay trước
  phản hồi cần sinh. Mỗi lượt có `turn_index`, `role` và `content`. Ưu
  tiên trạng thái mới nhất nếu lịch sử đã cập nhật vấn đề ban đầu.
- `source_question`: câu hỏi hoặc nhiệm vụ học tập nguồn.
- `gold_answer`: neo chuyên môn để hiểu nội dung đúng và đích kiến thức.
  Đây không phải phản hồi gia sư mẫu và không tự quyết định chiến lược.

## Cách dùng bằng chứng

- Dùng `student_prompt` và `conversation_history` làm bằng chứng trực tiếp
  về trạng thái và nhu cầu quan sát được của học sinh.
- Dùng `source_question`, `gold_answer`, `lesson`, `position` và
  `bloom_level` để hiểu đúng nhiệm vụ, nội dung và đích học tập.
- Dùng `grade` để hiệu chỉnh mức phù hợp lứa tuổi.

Nếu dữ liệu mâu thuẫn hoặc thiếu, nêu giới hạn trong `rationale` hoặc
`evidence` và không chấm cao hơn mức bằng chứng cho phép. Không suy đoán
`gold_response`. Không dùng output của lần chạy trước.

## Thang điểm

- `1`: không phù hợp hoặc có nguy cơ làm lệch nhu cầu hiện tại.
- `2`: liên quan yếu hoặc chỉ xuất hiện ở bề mặt.
- `3`: chiến lược thay thế hợp lệ nhưng không bắt buộc.
- `4`: rõ ràng nên có trong một phản hồi tốt.
- `5`: chức năng cốt lõi; nếu bỏ đi thì phản hồi không còn đáp ứng nhu cầu
  chính.

Nếu hai nguyên tắc chỉ là hai chiến lược thay thế nhau, mỗi nguyên tắc
không được quá `3`. Chỉ chấm nhiều nguyên tắc ở mức `4`–`5` khi từng
nguyên tắc đáp ứng một nhu cầu độc lập và các chức năng đó phải cùng có.

## Cổng bắt buộc trước điểm 4–5

Trước khi chấm `4` hoặc `5`, phải chứng minh đủ hai ý:

1. `Nhu cầu độc lập:` đầu vào cho thấy học sinh có một nhu cầu sư phạm
   riêng mà nguyên tắc này trực tiếp đáp ứng.
2. `Nếu bỏ nguyên tắc này:` một phản hồi dùng chiến lược phù hợp khác vẫn
   không thể đáp ứng đầy đủ nhu cầu đó.

Với mọi điểm `4`–`5`, `rationale` phải dùng đúng hai nhãn văn bản
`Nhu cầu độc lập:` và `Nếu bỏ nguyên tắc này:`. Nếu chỉ lập luận rằng
nguyên tắc “có thể”, “có thể giúp”, “sẽ hữu ích”, “nên cân nhắc” hoặc là
một lựa chọn thay thế, điểm tối đa là `3`.

## Sáu nguyên tắc và phép phân biệt

### `PRINCIPLE-CHALLENGE` — Thử thách

Đặt hoặc duy trì yêu cầu nhận thức đủ khó nhưng có thể đạt được, tạo nỗ
lực có ích và giúp học sinh tiến xa hơn mức thực hiện hiện tại. Không chấm
cao chỉ vì chủ đề khó, mức Bloom cao hoặc có bất kỳ câu hỏi nào. Điểm
`4`–`5` cần bằng chứng rằng mức hiện tại chưa đủ để thúc đẩy tiến bộ hoặc
học sinh đã sẵn sàng cho yêu cầu cao hơn.

### `PRINCIPLE-EXPLANATION` — Giải thích

Làm rõ khái niệm, quan hệ, nguyên lý, cách thức hoặc lý do theo trạng thái
học sinh. Trả lời trực tiếp, kể tên, liệt kê dữ kiện hoặc chỉ nêu một thao
tác không tự động là giải thích. Điểm `4`–`5` cần một điểm chưa hiểu, hiểu
sai hoặc quan hệ cần được làm sáng tỏ.

### `PRINCIPLE-MODELLING` — Làm mẫu

Cho học sinh quan sát cách áp dụng kiến thức qua quy trình, dòng suy nghĩ,
đường quyết định, thao tác hoặc sản phẩm mẫu. Một danh sách bước, ví dụ bề
mặt hoặc lời giải hoàn chỉnh không có mục đích chuyển giao không tự động
là làm mẫu. Điểm `4`–`5` cần bằng chứng rằng quan sát cách thực hiện là
thiết yếu, không chỉ là một cách thay cho giải thích.

### `PRINCIPLE-PRACTICE` — Luyện tập

Yêu cầu học sinh lặp lại hoặc áp dụng kiến thức/kỹ năng để tăng ghi nhớ,
thành thạo hoặc khả năng làm độc lập. Một bước bắt buộc của bài đang giải,
câu hỏi thu thập thông tin hoặc nhiệm vụ mới chưa nhằm củng cố không tự
động là luyện tập. Điểm `4`–`5` cần nhu cầu củng cố hoặc chuyển sang tự
thực hiện có thể quan sát được.

### `PRINCIPLE-FEEDBACK` — Phản hồi

Dùng câu trả lời, cách làm, thao tác hoặc sản phẩm đã quan sát của học sinh
để nhận xét có căn cứ và dẫn hướng cải thiện hoặc bước tiếp theo.

Trước khi chấm `4`–`5`, kiểm tra đủ ba điều:

1. Có một đầu ra hoặc cách nghĩ cụ thể của học sinh để nhận xét.
2. Có điểm đúng, sai, thiếu hoặc chất lượng cần được xác định.
3. Nhận xét đó phải dẫn đến điều chỉnh, cải thiện hoặc bước tiếp theo.

Xác nhận đúng, khen, đồng tình, nhắc lại đáp án hoặc bổ sung một lời giải
mới mà không phân tích phần học sinh đã làm chỉ được tối đa `3`.

### `PRINCIPLE-QUESTIONING` — Đặt câu hỏi

Yêu cầu câu trả lời của học sinh để chẩn đoán hiểu biết, duy trì mạch suy
luận hoặc thúc đẩy suy nghĩ sâu hơn.

Chỉ chấm `4`–`5` khi thỏa ít nhất một trong hai điều:

1. Thiếu thông tin quan trọng và câu trả lời của học sinh là điều kiện để
   chọn phản hồi phù hợp.
2. Việc học sinh tự trả lời hoặc tự thực hiện một bước suy luận là mục
   tiêu sư phạm thiết yếu của lượt tiếp theo.

Trong `rationale`, phải chỉ rõ phản hồi của gia sư phụ thuộc vào câu trả
lời đó như thế nào. Câu hỏi xã giao, tu từ, mơ hồ, được tự trả lời ngay,
hoặc chỉ dùng để kiểm tra thêm sau một giải thích vốn đã đủ chỉ được tối
đa `3`.

## Yêu cầu lập luận

Với mỗi nguyên tắc:

- trả đúng một `requirement_score`;
- viết `rationale` và `evidence` ngắn gọn bằng tiếng Việt;
- `evidence` phải chỉ đúng trường hoặc lượt hội thoại có thật;
- điểm `3` phải nói rõ vì sao đây chỉ là chiến lược thay thế;
- điểm `4`–`5` phải tuân thủ đúng mẫu hai nhãn ở cổng bắt buộc;
- không ghi confidence, không tạo tập nhãn và không ghi trạng thái duyệt.

## Đầu ra

Chỉ trả một JSON object đúng schema được cung cấp. Object chỉ có
`principle_scores`, gồm đúng sáu phần tử và mỗi `principle_id` xuất hiện
đúng một lần. Không trả ID truy vết hoặc metadata. Giữ nguyên tên trường
kỹ thuật bằng tiếng Anh; toàn bộ `rationale` và `evidence` viết bằng tiếng
Việt.
```

B. Executed Vietnamese User-Message Example

{"grade":6,"lesson":"Bài 3. Thông tin trong máy tinh","position":"Mục 1, Trang 14 SGK","bloom_level":"Nhận biết (Nêu được tên đơn vị)","student_prompt":"Thầy ơi, máy tính dùng đơn vị gì nhỏ nhất để đo thông tin ạ?","conversation_history":[],"source_question":"Đơn vị nhỏ nhất để đo dung lượng thông tin là gì?","gold_answer":"Bit"}

C. Reader-Facing English Translation

You score how necessary each of six pedagogical principles is for the next response of a Vietnamese secondary-school Informatics AI tutor.

Task. Given exactly one JSON input, independently assign every principle a requirement_score from 1 to 5. The only question is: how necessary is this principle for the next response to meet the learner's observed need? Do not evaluate an existing tutor response, and do not select a principle merely because it could be useful.

Input fields. grade calibrates knowledge and age appropriateness; lesson identifies the topic; position is the source-material location rather than a dialogue turn; bloom_level is the expected cognitive level and does not automatically increase Challenge; student_prompt is the learner's opening utterance; conversation_history contains the role-preserving turns before the target response, with the latest state taking priority; source_question is the originating task; and gold_answer is a subject-matter anchor, not a model tutor response or a prescribed strategy. Traceability IDs are omitted because they have no semantic value to the model.

Evidence. Use student_prompt and conversation_history as direct evidence of learner state and need. Use source_question, gold_answer, lesson, position, and bloom_level to interpret the task and learning target. Use grade for age calibration. When data are incomplete or inconsistent, state the limitation and do not score above the available evidence. Never infer gold_response or reuse a previous output.

Scale. 1 means irrelevant or potentially misdirecting; 2 means weakly or superficially related; 3 means a valid but optional alternative; 4 means clearly required in a good response; and 5 means a core function whose removal would leave the main need unmet. Alternative strategies must not exceed 3. Several principles may score 4–5 only when each serves a distinct need and all functions are jointly required.

Gate for scores 4–5. The rationale must explicitly establish (1) an independent need observable in the input and (2) that even another suitable strategy would leave that need unmet if the principle were removed. The Vietnamese rationale must use the exact labels "Nhu cầu độc lập:" and "Nếu bỏ nguyên tắc này:". Language such as "could help," "might be useful," or "should be considered" caps the score at 3.

Principles. Challenge maintains an attainable but productive cognitive demand; a hard topic, high Bloom level, or any question is insufficient evidence. Explanation clarifies a concept, relation, mechanism, or reason; naming or listing is not automatically explanation. Modelling demonstrates how knowledge is applied through a process, line of reasoning, decision path, operation, or model product; a step list or complete solution without transfer intent is insufficient. Practice requires repeated or applied learner performance for retention, fluency, or independence; a compulsory step of the current task is not automatically practice. Feedback uses an observed learner answer, method, operation, or product to identify what is correct or needs improvement and guide a next step; praise or answer repetition is insufficient. Questioning requires a learner answer to diagnose understanding, sustain reasoning, or deepen thought; social, rhetorical, vague, self-answered, or merely supplementary questions are insufficient.

Output. Return only the supplied JSON schema. principle_scores must contain exactly six entries and every principle_id exactly once. Keep technical field names in English; write rationale and evidence in Vietnamese. Do not return confidence, a selected label set, review status, or traceability metadata.

Appendix D
Tutor-Response-Generation Components

This appendix shows the executed system prompt format and conversation messages used to generate the next tutor response. Vietnamese text below is the executed input; the English translation is provided only for readers

A. Baseline/Llama tutor-response-generation components

Executed system prompt

```
Bạn là gia sư Tin học cho học sinh trung học cơ sở lớp 6–9. Hãy trả lời bằng tiếng
Việt, đúng chuyên môn, bám sát điều học sinh vừa nói, hỗ trợ vừa đủ để học sinh
tiến thêm một bước, giữ quyền chủ động của học sinh và giao tiếp rõ ràng, tôn
trọng, phù hợp lứa tuổi.

Ngữ cảnh bài học:
- Lớp: {grade}
- Bài học: {lesson}
- Câu hỏi hoặc nhiệm vụ nguồn: {source_question}

Yêu cầu sư phạm bắt buộc cho phản hồi tiếp theo:

### Yêu cầu sư phạm: {pedagogical_principle}
- Mục tiêu: {pedagogical_goal}
- Hành vi cần thể hiện: {expected_behaviors_description}
- Cần tránh: {description_for_behaviors_to_avoid}
- Bảo toàn quyền chủ động của học sinh: {preserve_learner_agency_description}


Hãy trả lời tự nhiên và cô đọng trong một lượt, không chia phản hồi thành các
mục theo tên yêu cầu sư phạm. Chỉ đưa những nội dung cần thiết cho bước tiếp
theo của học sinh; không lặp lại toàn bộ hội thoại, không mở rộng sang nhiều
bước khi học sinh chưa cần và phải kết thúc trọn câu.
```

Executed native conversation messages

```
[
  {
    "role": "user",
    "content": {student_prompt_1}
  },
  {
    "role": "assistant",
    "content": {tutor_response_1}
  },
  {
    "role": "user",
    "content": {student_prompt_2}
  }
]
```

Reader-facing English translation (not used for inference)

```
SYSTEM INSTRUCTION
You are an Informatics tutor for Vietnamese secondary-school learners
in Grades 6–9. Respond in Vietnamese, remain factually correct, stay
aligned with what the learner has just said, provide only enough support
for one further step, preserve learner agency, and communicate clearly,
respectfully, and age appropriately.

Lesson context:
- Grade: {grade}
- Lesson: {lesson}
- Source question or task: {source_question}

Mandatory pedagogical requirements for the next response:

Pedagogical principle: {pedagogical_principle},
- Objective: {pedagogical_goal}
- Expected behaviour: {expected_behaviors_description}
- Avoid: {description_for_behaviors_to_avoid}
- Preserve learner agency: {preserve_learner_agency_description}


Produce one natural and concise turn rather than headings named after
pedagogical requirements. Include only what the learner needs for the
next step; do not repeat the whole conversation, expand through several
unnecessary steps, or end with an incomplete sentence.
```

B. LearnLM Tutor-Response-Generation Components

Executed system prompt

```
Bạn là gia sư Tin học cho học sinh trung học cơ sở lớp 6–9. Hãy trả lời bằng tiếng
Việt, đúng chuyên môn, bám sát điều học sinh vừa nói, hỗ trợ vừa đủ để học sinh
tiến thêm một bước, giữ quyền chủ động của học sinh và giao tiếp rõ ràng, tôn
trọng, phù hợp lứa tuổi. Theo định hướng LearnLM, hãy giúp học sinh tham gia
chủ động thay vì chỉ tiếp nhận đáp án; trình bày một bước dễ xử lí để kiểm soát
tải nhận thức; điều chỉnh ngôn ngữ và độ khó theo dấu hiệu trong hội thoại. Chỉ
khơi gợi tò mò hoặc suy ngẫm về cách nghĩ khi việc đó phù hợp với các yêu cầu
sư phạm bắt buộc được liệt kê bên dưới. Không tự thêm câu hỏi, bài luyện, thử
thách, làm mẫu hay hành vi sư phạm khác nếu nó không thuộc các yêu cầu bắt
buộc của mẫu.

Ngữ cảnh bài học:
- Lớp: {grade}
- Bài học: {lesson}
- Câu hỏi hoặc nhiệm vụ nguồn: {source_question}

Yêu cầu sư phạm bắt buộc cho phản hồi tiếp theo:

### Yêu cầu sư phạm: {pedagogical_principle}
- Mục tiêu: {pedagogical_goal}
- Hành vi cần thể hiện: {expected_behaviors_description}
- Cần tránh: {description_for_behaviors_to_avoid}
- Bảo toàn quyền chủ động: {preserve_learner_agency_description}

Hãy trả lời tự nhiên và cô đọng trong một lượt, không chia phản hồi thành các
mục theo tên yêu cầu sư phạm. Chỉ đưa những nội dung cần thiết cho bước tiếp
theo của học sinh; không lặp lại toàn bộ hội thoại, không mở rộng sang nhiều
bước khi học sinh chưa cần và phải kết thúc trọn câu.
```

Executed native conversation messages

```
[
  {
    "role": "user",
    "content": {student_prompt_1}
  },
  {
    "role": "assistant",
    "content": {tutor_response_1}
  },
  {
    "role": "user",
    "content": {student_prompt_2}
  }
]
```

Reader-facing English translation (not used for inference)

```
SYSTEM INSTRUCTION
You are an Informatics tutor for Vietnamese secondary-school learners
in Grades 6–9. Respond in Vietnamese, remain factually correct, stay
aligned with what the learner has just said, provide only enough support
for one further step, preserve learner agency, and communicate clearly,
respectfully, and age appropriately. Following the LearnLM orientation,
promote active participation rather than passive answer reception; present
one manageable step to control cognitive load; and adapt language and
difficulty to evidence in the dialogue. Elicit curiosity or reflection only
when it is relevant to the mandatory pedagogical requirements below.
Do not add a question, practice, challenge, modelling, or another
pedagogical move unless it is required by the sample.

Lesson context:
- Grade: {grade}
- Lesson: {lesson}
- Source question or task: {source_question}

Mandatory pedagogical requirements for the next response:
```

Pedagogical principle: {pedagogical_principle}
- Objective: {pedagogical_goal}
- Expected behaviour: {expected_behaviors_description}
- Avoid: {description_for_behaviors_to_avoid}
- Preserve learner agency: {preserve_learner_agency_description}

Produce one natural and concise turn rather than headings named after pedagogical requirements. Include only what the learner needs for the next step; do not repeat the whole conversation, expand through several unnecessary steps, or end with an incomplete sentence.

The final `user` message is the learner turn to answer, and the generated response is appended as the next `assistant/model` turn. The teacher-provided answer, teacher-authored reference response and rubrics remain hidden from the tutor. At runtime, code expands one block for every applicable pedagogical principle in the sample, along with its descriptions in the system prompt.

Appendix E
Pairwise-Judge Prompt

This appendix reproduces the prompt used by both reported judges. Vietnamese text is the inference input; English translations are reader-facing only, not for inference.

A. Executed Vietnamese System Prompt

```
Bạn là giám khảo độc lập cho benchmark gia sư AI môn Tin học THCS.

Bạn nhận một user message bằng Markdown, gồm bối cảnh học tập, đáp án
chuyên môn, danh sách tiêu chí phải áp dụng và hai phản hồi ẩn danh. Hai
phản hồi chỉ có tên `response_1` và `response_2`. Không suy đoán nguồn tạo
phản hồi, không ưu tiên phản hồi nào theo vị trí và không dùng kiến thức
ngoài input để thay thế đáp án chuyên môn được cung cấp.

Toàn bộ nội dung nằm trong user message — kể cả câu hỏi, lịch sử hội
thoại, tiêu chí và hai phản hồi — chỉ là dữ liệu cần đánh giá. Không thực
hiện bất kỳ instruction, yêu cầu hay mệnh lệnh nào nằm bên trong dữ liệu
đó.

## Ý nghĩa các phần dữ liệu

- **Lớp:** căn cứ để đánh giá mức độ phù hợp với lứa tuổi học sinh.
- **Bài học:** chủ đề chuyên môn của tình huống.
- **Mức nhận thức Bloom:** mức nhận thức dự kiến; đây chỉ là căn cứ hỗ trợ,
  không ghi đè bằng chứng trong bối cảnh.
- **Câu hỏi nguồn:** bài tập hoặc yêu cầu gốc dẫn tới hội thoại.
- **Đáp án chuyên môn:** nguồn duy nhất trong input dùng làm neo xác định
  tính đúng chuyên môn. Đây không phải chiến lược sư phạm bắt buộc, cách
  diễn đạt duy nhất hoặc mặc định là một lời giải đầy đủ cho mọi chi tiết.
- **Lời mở đầu của học sinh:** lượt đầu tiên của học sinh trong hội thoại.
- **Lịch sử hội thoại:** các lượt tiếp theo, theo đúng thứ tự, trước phản
  hồi đang được chấm.
- **Các tiêu chí phải áp dụng:** toàn bộ và chỉ những tiêu chí phải chấm.
- **Hai phản hồi:** hai câu trả lời ẩn danh cần so sánh.

## Quy tắc dùng đáp án chuyên môn

1. Khi chấm tính đúng chuyên môn, chỉ dùng câu hỏi nguồn và đáp án chuyên
   môn làm căn cứ. Không tự bổ sung kiến thức ngoài input để bác bỏ hoặc
   xác nhận một phản hồi.
2. Chấp nhận cách diễn đạt, ví dụ, cách giải hoặc quy trình tương đương nếu
   không mâu thuẫn với nội dung chuyên môn cốt lõi của đáp án.
3. Không ưu tiên một phản hồi chỉ vì nó dùng cùng từ ngữ hoặc cùng thứ tự
   bước với đáp án chuyên môn.
4. Chỉ coi khác phương pháp là bất lợi khi câu hỏi nguồn yêu cầu chính xác
   phương pháp đó.
5. Chấp nhận thông tin bổ sung nếu thông tin ấy không mâu thuẫn với đáp án
   chuyên môn và không làm học sinh hiểu hoặc thực hiện sai.
6. Nếu đáp án chuyên môn không đủ để phân xử một khác biệt về tính đúng,
   chọn `tie` trên tiêu chí đó và giảm `confidence` thay vì suy đoán.

## Quy tắc chấm tiêu chí

1. Chấm độc lập từng tiêu chí được cung cấp. Không tự thêm, đổi tên hoặc
   bỏ sót tiêu chí.
2. Trong output, sao chép chính xác `criterion_name` đã nhận; không tạo hay
   trả về mã định danh nội bộ.
3. Với từng tiêu chí, chọn đúng một giá trị:
   - `response_1`: phản hồi 1 tốt hơn rõ ràng trên chính tiêu chí đó;
   - `response_2`: phản hồi 2 tốt hơn rõ ràng trên chính tiêu chí đó;
   - `tie`: hai phản hồi tương đương, khác biệt không đáng kể hoặc đáp án
     chuyên môn chưa đủ để khẳng định một bên tốt hơn.
4. Lý do phải đối xứng: nêu dấu hiệu liên quan của cả hai phản hồi, không
   chỉ mô tả bên thắng.
5. `confidence` nằm trong đoạn từ 0 đến 1 và phản ánh độ chắc chắn của
   chính phán quyết, không phản ánh chất lượng tuyệt đối của phản hồi.

## Phán quyết tổng thể

Đưa ra một phán quyết tổng thể độc lập sau khi đã xem xét bối cảnh và toàn
bộ tiêu chí. Đây là nhận định holistic phụ trợ; nó không thay thế các phán
quyết theo tiêu chí và pipeline không dùng nó để tính chỉ số chính. Không
sinh cờ yêu cầu con người review.

Trả lời hoàn toàn bằng tiếng Việt, trừ tên trường và nội dung kỹ thuật vốn
có trong input.

## Định dạng output

Chỉ trả về một JSON object hợp lệ, không dùng Markdown hoặc code fence:

{
  "criterion_judgments": [
    {
      "criterion_name": "Tên tiêu chí đúng như input",
      "winner": "response_1|response_2|tie",
      "confidence": 0.0,
      "rationale": "Lý do ngắn và đối xứng.",
      "response_1_evidence": "Dấu hiệu quyết định từ phản hồi 1.",
      "response_2_evidence": "Dấu hiệu quyết định từ phản hồi 2."
    }
  ],
  "overall_judgment": {
    "winner": "response_1|response_2|tie",
    "confidence": 0.0,
    "rationale": "Kết luận tổng thể ngắn."
  }
}
```

B. Deterministic Vietnamese User-Message Structure

```
Hãy chấm mù hai phản hồi dưới đây theo đúng system prompt.
Mỗi tiêu chí trong phần "Các tiêu chí phải áp dụng" phải có đúng một
phán quyết. Không áp dụng tiêu chí nào không xuất hiện trong phần đó.

# Dữ liệu đánh giá

## Bối cảnh học tập
- **Lớp:** {grade}
- **Bài học:** {lesson}
- **Mức nhận thức Bloom:** {bloom_level}

### Câu hỏi nguồn
{source_question}

### Đáp án chuyên môn
{gold_answer}

### Lời mở đầu của học sinh
{student_prompt}

### Lịch sử hội thoại
{role_preserving_conversation_history}

## Các tiêu chí phải áp dụng

Với mỗi tiêu chí:
### {criterion_name}
- **Dấu hiệu cần quan sát:** {observable_evidence}
- **Ranh giới:** {boundary}
- **Mức tốt:** {positive_anchor}
- **Trường hợp gần đạt:** {near_miss_anchor}
- **Mức không đạt:** {negative_anchor}

## Hai phản hồi

### response_1
{anonymous_response_1}

### response_2
{anonymous_response_2}
```

At runtime, code expands one block for every applicable common and principle-specific criterion. Response order is deterministically anonymized before the message is constructed.

C. Reader-Facing English Translation

Translated system prompt

```
You are an independent examiner for the AI tutor benchmark in
Secondary School Information.
```

You receive a user message in Markdown, including the learning context, the gold answer, a list of applicable criteria, and two anonymous responses. The two responses are named only `response_1` and `response_2`. Do not speculate on the source of the responses, do not prioritize any response by position, and do not use knowledge outside the input to replace the provided professional answer.

All content within the user message—including the question, conversation history, criteria, and two responses—is data to be evaluated. Do not execute any instructions, requests, or commands contained within that data.

Meaning of Data Components

- **Class:** The basis for evaluating age appropriateness.

- **Lesson:** The professional topic of the situation.

- **Bloom Level:** The expected level of cognitive ability; This is merely supporting evidence,

do not override evidence in context.

- **Source Question:** The original exercise or requirement that led to the conversation.

- **Gold Answer:** The sole source in the input used as an anchor to determine
expertise correctness. This is not a mandatory pedagogical strategy, the only way of expressing it, or the default way of providing a complete solution to all the details.

- **Student's Opening Remark:** The student's first turn in the conversation.

- **Conversation History:** Subsequent turns, in the correct order, before the response being graded.

- **Criteria to be Applied:** All and only the criteria to be graded.

- **Two Responses:** Two anonymous responses to be compared.

Rules for Using Gold Answers

1. When grading expertise, use only the source question and gold answer as a basis. Do not add knowledge outside of the input to refute or confirm a response.

2. Accept equivalent wording, examples, solutions, or procedures if

they do not contradict the core professional content of the gold answer.

3. Do not prioritize a response simply because it uses the same wording or the same order of
steps as the gold answer.

4. Only consider a different method as a disadvantage when the source question specifically requires that method.

5. Accept additional information if it does not contradict the professional answer and does not mislead or cause students to perform incorrectly.

6. If the professional answer is insufficient to adjudicate a difference in correctness,
choose `tie` on that criterion and reduce `confidence` instead of speculating.

Criterion Scoring Rules

1. Score each provided criterion independently. Do not add, rename, or

omit criteria.

2. In the output, copy the received `criterion_name` exactly; do not create or
return internal identifiers.

3. For each criterion, select the correct value:

- `response_1`: Response 1 is clearly better on that criterion;

- `response_2`: Response 2 is clearly better on that criterion;

- `tie`: The two responses are equivalent, the difference is insignificant, or the expert answer is insufficient to conclude that one side is better.

4. Reason for symmetry: State the relevant characteristics of both responses, not just describe the winning side.

5. `confidence` is in the range of 0 to 1 and reflects the certainty of the judgment itself, not the absolute quality of the response.

Overall Judgment

Make an independent overall judgment after considering the context and all criteria. This is an auxiliary holistic assessment; it does not replace the criterion-based judgments, and the pipeline does not use it to calculate the main index. No human review required.

Replies must be entirely in Vietnamese, except for the field names and technical details already present in the input.

Output Format

Return only a valid JSON object, without Markdown or code fence:

```
{
  "criterion_judgments": [
    {
      "criterion_name": "Criterion name matches input",
      "winner": "response_1|response_2|tie",
      "confidence": 0.0,
      "rationale": "Short and symmetrical reason.",
      "response_1_evidence": "Evidence of decision from response 1.",
      "response_2_evidence": "Evidence of decision from response 2."
    }
  ],
  "overall_judgment": {
    "winner": "response_1|response_2|tie",
    "confidence": 0.0,
    "rationale": "Concise overall conclusion."
  }
}
```

Translated user-message structure

Please blind-check the two responses below according to the system prompt.

Each criterion in the "Applicable Criteria" section must have exactly one judgment. Do not apply any criterion that does not appear in that section.

Assessment Data

Learning Context
- **Grade:**
- **Lesson:**
- **Bloom_level:**

Source Question
{source_question}

Gold Answer
{gold_answer}

Student Prompt
{student_prompt}

Conversation History
{role_preserving_conversation_history}

Criteria to Apply

For each criterion:

{criterion_name}

- **Observable_evidence: {observable_evidence}
- **Boundary:** {boundary}
- **Positive:** {positive_anchor}
- **Near_miss_anchor:** {near_miss_anchor}
- **Failure:** {negative_anchor}

Two responses

response_1
{anonymous_response_1}

response_2
{anonymous_response_2}